

Assignment 5 — Math Fundamentals for Robotics 16-811, Fall 2024

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1. Consider a plane curve $y(x)$ over the interval $[x_0, x_1]$, with specified endpoints $y_0 = y(x_0)$ and $y_1 = y(x_1)$. Assume that $y_0 > 0$ and $y_1 > 0$ and that $y(x) \geq 0$ for $x_0 \leq x \leq x_1$. Now imagine rotating the curve about the x-axis to obtain a surface of revolution. Find the C^2 curve $y(x)$ with specified endpoints that minimizes the surface area of this surface of revolution.

[Hint: This problem explores further some of the limitations of the Calculus of Variations. Depending on the endpoint conditions there may or may not be a C^2 solution. What does the optimal “curve” look like when it is not C^2 ? Can you say how the endpoint conditions matter? Be aware: There are many subtleties; don’t expect to cover all, but explore what you can.]

Solution:

Surface area of revolution is given by

$$S = \int_{x_0}^{x_1} 2\pi y(x) \sqrt{1 + (y'(x))^2} dx.$$

We want to minimize S subject to the constraints $y(x_0) = y_0, y(x_1) = y_1$, so the optimization program is

$$\begin{aligned} \min_y J(y) &= \int_{x_0}^{x_1} F(x, y, y') dx, \\ \text{s.t. } &\begin{cases} y(x_0) &= y_0 \\ y(x_1) &= y_1 \end{cases} \end{aligned}$$

Where $F(x, y, y') = 2\pi y(x) \sqrt{1 + (y'(x))^2}$.

Now suppose y^* minimizes J , and $\eta \in C^2[x_0, x_1]$ be a random Δ function s.t. $\eta(x_0) = \eta(x_1) = 0$. And we define $y(x) = y^*(x) + \epsilon \eta(x)$, where $\epsilon \in \mathbb{R}$ is a small number. Then we can define $J(S)$ w.r.t. ϵ as

$$\begin{aligned} \Phi(\epsilon) &= \int_{x_0}^{x_1} F(x, y, y') dx \\ &= \int_{x_0}^{x_1} F(x, y^*(x) + \epsilon \eta(x), y^{*'}(x) + \epsilon \eta'(x)) dx. \end{aligned}$$

And we can calculate $\Phi'(0)$ as

$$\begin{aligned} \Phi'(0) &= \int_{x_0}^{x_1} \frac{\partial F}{\partial y} \eta(x) dx + \int_{x_0}^{x_1} \frac{\partial F}{\partial y'} \eta'(x) dx \\ &= \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx. \end{aligned}$$

Because y^* minimizes J , we have $\Phi'(0) = 0$. Also because η is arbitrary, so we must have the Euler-Lagrange equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

Because we have

$$\begin{cases} \frac{\partial F}{\partial y} &= 2\pi \sqrt{1 + (y'(x))^2} \\ \frac{\partial F}{\partial y'} &= 2\pi y(x) \frac{y'(x)}{\sqrt{1 + (y'(x))^2}} \end{cases}$$

Also because $F(x, y(x), y'(x))$ does not explicitly depend on x , we can use the **Beltrami Identity** to simplify the equation as

$$F - y'(x) \frac{\partial F}{\partial y'} = C = \text{constant} \Rightarrow$$

$$2\pi y(x) \sqrt{1 + (y'(x))^2} - 2\pi y'(x) \frac{y(x)y'(x)}{\sqrt{1 + (y'(x))^2}} = 2\pi \frac{y(x)}{\sqrt{1 + (y'(x))^2}} = C \Rightarrow$$

$$y'(x) = \pm \sqrt{(Cy)^2 - 1}$$

Note that the constant C is absorbing all the constants in the equation, so we can solve the equation by separating variables and integrating both sides as

$$\int \frac{dy}{\sqrt{(Cy)^2 - 1}} = \pm \int dx.$$

Then we get

$$y(x) = C_1 \cosh \left(\frac{x - C_2}{C_1} \right)$$

Where constants C_1 and C_2 depends on the boundary conditions (x_0, y_0) and (x_1, y_1) .

The shape visualized with Desmos looks like a thin vase with big top and bottom, which is reasonable since we want the shape to shrink a little before it reaches another boundary, in order to minimize the surface area.

2. Using Calculus of Variations, show that the shortest curve between two points on a sphere is an arc of a great circle. [Hints: There are different ways to solve this problem. Here is one approach: Use spherical (u, v) coordinates, where $x = R \sin v \cos u, y = R \sin v \sin u, z = R \cos v$, with R the radius of the sphere. (Do not worry about singularities in the representation.) Rephrase 3D arclength $\sqrt{dx^2 + dy^2 + dz^2}$ as a function of u, v, du, dv . View the curve as a function $v(u)$. Observe that u does not appear directly in the resulting integrand for arclength, so replace the Euler-Lagrange equation with another, as in lecture. You may find the following identity useful:

$$\int \frac{adw}{\sqrt{\sin^4 w - a^2 \sin^2 w}} = -\sin^{-1}\left(\frac{\cot w}{\sqrt{\frac{1}{a^2} - 1}}\right) + k,$$

where a and k are appropriate constants.]

Solution:

The arclength of a curve on a sphere is given by

$$L = \int_{u_0}^{u_1} R \sqrt{[v'(u)]^2 + \sin^2 v(u)} du.$$

Now remove constant linear parameter R and let $F(u, v(u), v'(u)) = \sqrt{[v'(u)]^2 + \sin^2 v(u)}$. The optimization program is formally constructed as

$$\begin{aligned} \min_v J(v) &= \int_{u_0}^{u_1} F(u, v(u), v'(u)) du, \\ \text{s.t. } \begin{cases} v(u_0) &= v_0 \\ v(u_1) &= v_1 \end{cases} \end{aligned}$$

Omitting all the derivation we did in Question 1, the Euler-Lagrange equation is

$$\frac{\partial F}{\partial v} - \frac{d}{du} \left(\frac{\partial F}{\partial v'} \right) = 0.$$

And we have

$$\begin{cases} \frac{\partial F}{\partial v} &= \frac{\sin v(u) \cos v(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} \\ \frac{\partial F}{\partial v'} &= \frac{v'(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} \end{cases}$$

Similar to Q1, because $F(u, v(u), v'(u))$ does not explicitly depends on u , we can apply the **Beltrami Identity** to simplify the Euler-Lagrange equation as

$$\begin{aligned} F - v'(u) \frac{\partial F}{\partial v'} &= C = \text{constant} \Rightarrow \\ \sqrt{[v'(u)]^2 + \sin^2 v(u)} - v'(u) \frac{v'(u)}{\sqrt{[v'(u)]^2 + \sin^2 v(u)}} &= C \Rightarrow \\ \sin^2 v(u) &= C \sqrt{[v'(u)]^2 + \sin^2 v(u)} \Rightarrow \\ [v'(u)]^2 &= \frac{\sin^4 v(u)}{C^2} - \sin^2 v(u) \Rightarrow \\ v'(u) &= \pm \sqrt{\frac{\sin^4 v(u)}{C^2} - \sin^2 v(u)} \end{aligned}$$

Separate variables and integrate both sides, then we can get

$$\int \frac{C^2 dv(u)}{\sqrt{\sin^4 v(u) - C^2 \sin^2 v(u)}} = \pm \int du.$$

Using the identity from hint, we have

$$\begin{aligned}
 -\sin^{-1}\left(\frac{\cot v(u)}{\sqrt{\frac{1}{C^2}-1}}\right) &= \pm u + C' \Rightarrow \\
 \cot v(u) &= -\sin(\pm u + C')\sqrt{\frac{1}{C^2}-1} \Rightarrow \\
 \tan v(u) &= -\frac{\frac{C}{\sqrt{1-C^2}}}{\sin(\pm u + C')}
 \end{aligned}$$

Absorbing the signs and rearranging the constants, we can get

$$v(u) = \arctan\left(\frac{C_1}{\sin(u + C_2)}\right)$$

Where C_1, C_2 are constants depending on the boundary conditions (u_0, v_0) and (u_1, v_1) .

The shape visualized with Desmos looks like a periodic function with a period of 2π , where C_1 controls amplitude C_2 controls phase. This is reasonable since u is the longitude angle ranging $[0, 2\pi]$, and v is the latitude angle ranging $[0, \pi]$, so the curve should be periodic with a period of 2π since it's a function of u .

3. In the brachistochrone problem (“ski-slope” in lecture), suppose the right endpoint (x_1, y_1) is not specified exactly, but is merely constrained to satisfy an equation of the form $g(x, y) = 0$. Show that the optimizing curve $y(x)$ is perpendicular to the iso-contour $g(x, y) = 0$ at (x_1, y_1) .

[Hints: (i) Use an equation from lecture rather than derive everything from scratch. (ii) Observe that the curves are written differently, one explicitly as $y(x)$, the other implicitly as $g(x, y) = 0$. Think about how to compare the slopes of the two curves despite these different representations.]

Solution:

Similar to what was taught in the lecture, let s denote arclength and let $v = \frac{ds}{dt}$ denote instant linear velocity along the curve, then the time to reach (x_1, y_1) is given by

$$\begin{aligned} J(y) &= \int_{x_0}^{x_1} \frac{ds}{v} \\ &= \int_{x_0}^{x_1} \frac{\sqrt{1 + (y'(x))^2}}{v} dx \\ &= \frac{1}{\sqrt{-2g}} \int_{x_0}^{x_1} \frac{\sqrt{1 + (y'(x))^2}}{\sqrt{y_0 - y(x)}} dx \end{aligned}$$

So formally, the optimization program is

$$\begin{aligned} \min_y J(y) &= \int_{x_0}^{x_1} F(x, y(x), y'(x)) dx, \\ \text{s.t. } \begin{cases} y(x_0) &= y_0 \\ g(x_1, y_1) &= 0 \end{cases} \end{aligned}$$

Where $F(x, y(x), y'(x)) = \frac{\sqrt{1 + (y'(x))^2}}{\sqrt{y_0 - y(x)}}$.

Now suppose y^* minimizes J , and $\eta \in C^2[x_0, x_1]$ be a random Δ function s.t. $\eta(x_0) = 0$, we can define the variation $y(x) = y^*(x) + \epsilon\eta(x)$, where $\epsilon \in \mathbb{R}$ is a small number. Then we can define $J(y)$ w.r.t. ϵ as

$$\begin{aligned} \Phi(\epsilon) &= \int_{x_0}^{x_1(\epsilon)} F(x, y, y') dx \\ &= \int_{x_0}^{x_1(\epsilon)} F(x, y^*(x) + \epsilon\eta(x), y^{*'}(x) + \epsilon\eta'(x)) dx. \end{aligned}$$

And we can calculate $\Phi'(0)$ as

$$\begin{aligned} \Phi'(0) &= \left. \frac{dx_1}{d\epsilon} F \right|_{x_1(0)} + \int_{x_0}^{x_1(0)} \left[\frac{\partial F}{\partial y} \eta(x) - \frac{\partial F}{\partial y'} \eta'(x) \right] dx \\ &= \left(\frac{dx_1}{d\epsilon} F + \frac{\partial F}{\partial y'} \eta(x) \right) \Big|_{x_1(0)} + \int_{x_0}^{x_1(0)} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx \end{aligned}$$

Because the right endpoint is constrained to only satisfy $g(x_1, y_1) = 0$, we can define the constraint to be $g(x_1(\epsilon), y^*(x_1(\epsilon)) + \epsilon\eta(x_1(\epsilon))) = 0$ and make $g'(0) = 0$ at the right endpoint

$$\begin{aligned} g'(0) &= \left. \frac{d}{d\epsilon} g(x_1(\epsilon), y^*(x_1(\epsilon)) + \epsilon\eta(x_1(\epsilon))) \right|_{\epsilon=0} = 0 \Rightarrow \\ g_x \frac{dx_1}{d\epsilon} + g_y y^{*'}(x_1(\epsilon)) \frac{dx_1}{d\epsilon} + g_y \eta(x_1(\epsilon)) + \epsilon g_y \eta'(x_1(\epsilon)) \frac{dx_1}{d\epsilon} \Big|_{\epsilon=0} &= 0 \Rightarrow \\ g_x \frac{dx_1}{d\epsilon} + g_y y^{*'}(x_1(0)) \frac{dx_1}{d\epsilon} + g_y \eta(x_1(0)) &= 0 \Rightarrow \\ \frac{dx_1}{d\epsilon} &= - \frac{g_y \eta(x_1(0))}{g_x + g_y y^{*'}(x_1(0))} \end{aligned}$$

Substitute $\frac{dx_1}{d\epsilon}$ back to $\Phi'(0)$, we have

$$\begin{aligned}\Phi'(0) &= \left(-\frac{g_y \eta(x_1(0))}{g_x + g_y (y^*)'(x_1(0))} F + \frac{\partial F}{\partial y'} \eta(x) \right) \Big|_{x_1(0)} + \int_{x_0}^{x_1(0)} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx \\ &= \left(-\frac{g_y}{g_x + g_y y^{*'}(x_1(0))} F + \frac{\partial F}{\partial y'} \right) \eta(x_1(0)) + \int_{x_0}^{x_1(0)} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) dx\end{aligned}$$

In order to make $\Phi'(0) = 0$ for arbitrary $\eta(x)$, we must have both

$$\begin{cases} -\frac{g_y}{g_x + g_y y^{*'}(x_1(0))} F + \frac{\partial F}{\partial y'} = 0 \\ \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \end{cases}$$

Combine them together, we have

$$\begin{aligned}\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) &= \frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{g_y}{g_x + g_y y^{*'}(x_1(0))} F \right) = 0 \Rightarrow \\ \frac{y'(x)}{\sqrt{(y_0 - y(x))(1 + (y'(x))^2)}} - \frac{d}{dx} \left(\frac{g_y}{g_x + g_y y^{*'}(x_1(0))} \frac{\sqrt{1 + (y'(x))^2}}{\sqrt{y_0 - y(x)}} \right) &= 0\end{aligned}$$

omg this looks horrible, let's try another way.

Again similar to Q1 and Q2, since $F(x, y(x), y'(x))$ does not directly depend on x , we can apply the **Beltrami Identity** to simplify the Euler-Lagrange equation as

$$\begin{aligned}F - y'(x) \frac{\partial F}{\partial y'} &= C = \text{constant} \Rightarrow \\ F - y'(x) \left(\frac{g_y}{g_x + g_y y^{*'}(x_1(0))} \right) F &= C \Rightarrow\end{aligned}$$

If we looked at the right endpoint (x_1, y_1) , we have

$$\begin{aligned}\frac{g_x}{g_x + g_y y(x_1)} F \Big|_{x=x_1} &= C \Rightarrow \\ \frac{g_x}{g_x + g_y y(x_1)} \frac{\sqrt{1 + (y'(x_1))^2}}{\sqrt{y_0 - y(x_1)}} &= C \Rightarrow \\ \frac{g_x^2 (1 + [y'(x_1)]^2)}{(g_x^2 + 2g_x g_y y(x_1) + g_y^2 [y(x_1)]^2)(y_0 - y(x_1))} &= C^2\end{aligned}$$

omg this looks horrible again, let's try another way.

Let's examine the endpoint condition again, we have $g(x_1, y_1) = 0$, so the variations δx , δy are related by

$$g_x \delta x_1 + g_y \delta y_1 = 0 \Rightarrow \delta x_1 = -\frac{g_y}{g_x} \delta y_1$$

And from some ChatGPT conversations, we learned the transversality condition

$$\left[F - y' \frac{\partial F}{\partial y'} \right] \delta x_1 + \frac{\partial F}{\partial y'} \delta y_1 = 0$$

Substituting with the above equation, we have

$$\begin{aligned}\left[F - y' \frac{\partial F}{\partial y'} \right] \left(-\frac{g_y}{g_x} \delta y_1 \right) + \frac{\partial F}{\partial y'} \delta y_1 &= 0 \Rightarrow \\ \left(\frac{\partial F}{\partial y'} - \left[F - y' \frac{\partial F}{\partial y'} \right] \frac{g_y}{g_x} \right) \delta y_1 &= 0\end{aligned}$$

Since δy_1 is arbitrary, we must have

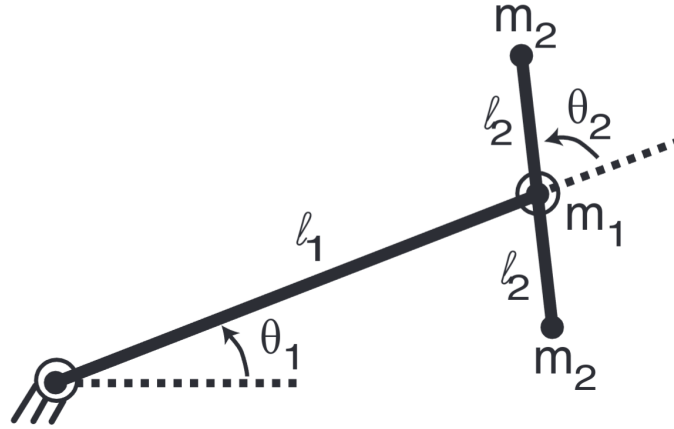
$$\frac{\partial F}{\partial y'} - \left[F - y' \frac{\partial F}{\partial y'} \right] \frac{g_y}{g_x} = 0$$

Substituting with $F - y' \frac{\partial F}{\partial y'} = C$, we have

$$y' C - C \left(\frac{g_y}{g_x} \right) = 0 \Rightarrow y' = \frac{g_y}{g_x}$$

Which exactly means that the optimizing curve $y(x)$ is perpendicular to the iso-contour $g(x, y) = 0$ at (x_1, y_1) .

- 4.(a) Using Lagrangian Dynamics, derive the relationship between joint torques and the angular state (angles, angular velocities, and angular accelerations) of the following balanced manipulator:



There is no gravity (in practice, gravity is perpendicular to the sheet of the paper).

Legend: All of link #1's mass, m_1 , is concentrated at distance l_1 from its rotational joint (which is attached to the ground). In turn, link #2 rotates around this distal point, with two masses, m_2 , located symmetrically, each at distance l_2 , from the joint. In practice, these two masses might constitute one counter-balanced end-effector or two different but equally weighted end-effectors. — This is a variation of a basic SCARA-type robot arm, used in industrial assembly.

Solution:

The velocity of each mass is given by

$$\begin{cases} \dot{x}_1 &= -l_1 \sin \theta_1 \dot{\theta}_1 \\ \dot{y}_1 &= l_1 \cos \theta_1 \dot{\theta}_1 \\ \dot{x}_2 &= -(l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2))\dot{\theta}_1 - l_2 \sin(\theta_1 + \theta_2)\dot{\theta}_2 \\ \dot{y}_2 &= (l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2))\dot{\theta}_1 + l_2 \cos(\theta_1 + \theta_2)\dot{\theta}_2 \\ \dot{x}_{2'} &= -(l_1 \sin \theta_1 - l_2 \sin(\theta_1 + \theta_2))\dot{\theta}_1 + l_2 \sin(\theta_1 + \theta_2)\dot{\theta}_2 \\ \dot{y}_{2'} &= (l_1 \cos \theta_1 - l_2 \cos(\theta_1 + \theta_2))\dot{\theta}_1 - l_2 \cos(\theta_1 + \theta_2)\dot{\theta}_2 \end{cases}$$

The Lagrangian of the system is given by

$$\begin{aligned} L &= T - V \\ &= T \\ &= \frac{1}{2}m_1(\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2}m_2(\dot{x}_2^2 + \dot{y}_2^2 + \dot{x}_{2'}^2 + \dot{y}_{2'}^2) \\ &= \frac{1}{2}m_1 l_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 \left[(l_1^2 + l_2^2 + 2l_1 l_2 \cos \theta_2) \dot{\theta}_1^2 + 2l_2 (l_1 \cos \theta_2 + l_2) \dot{\theta}_1 \dot{\theta}_2 + l_2^2 \dot{\theta}_2^2 \right] \\ &\quad + \left[(l_1^2 + l_2^2 - 2l_1 l_2 \cos \theta_2) \dot{\theta}_1^2 + 2l_2 (-l_1 \cos \theta_2 + l_2) \dot{\theta}_1 \dot{\theta}_2 + l_2^2 \dot{\theta}_2^2 \right] \\ &= \left(\frac{1}{2}m_1 l_1^2 + m_2 (l_1^2 + l_2^2) \right) \dot{\theta}_1^2 + 2m_2 l_2^2 \dot{\theta}_1 \dot{\theta}_2 + m_2 l_2^2 \dot{\theta}_2^2 \end{aligned}$$

So the relationship between joint torques can be derived from the Euler-Lagrange equation as

$$\begin{aligned} \tau_1 &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} \Rightarrow \\ \tau_1 &= \frac{d}{dt} \left((m_1 l_1^2 + 2m_2 (l_1^2 + l_2^2)) \dot{\theta}_1 + 2m_2 l_2^2 \dot{\theta}_2 \right) - 0 \Rightarrow \\ \tau_1 &= (m_1 l_1^2 + 2m_2 (l_1^2 + l_2^2)) \ddot{\theta}_1 + 2m_2 l_2^2 \ddot{\theta}_2 \end{aligned}$$

And

$$\begin{aligned}\tau_2 &= \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) - \frac{\partial L}{\partial \theta_2} \Rightarrow \\ \tau_2 &= \frac{d}{dt} \left(2m_2 l_2^2 \dot{\theta}_1 + 2m_2 l_2^2 \dot{\theta}_2 \right) - 0 \Rightarrow \\ \tau_2 &= 2m_2 l_2^2 \ddot{\theta}_1 + 2m_2 l_2^2 \ddot{\theta}_2\end{aligned}$$

- (b) When $\ddot{\theta}_2 = 0$, explain the terms relating $\ddot{\theta}_1$ to τ_1 .

Solution:

When $\ddot{\theta}_2 = 0$, the relationship between $\ddot{\theta}_1$ and τ_1 is given by

$$\tau_1 = (m_1 l_1^2 + 2m_2 (l_1^2 + l_2^2)) \ddot{\theta}_1$$

The system, which occurred in high school physics questions, can be viewed in a holistic way, basically recenter the mass according to m_1, m_2, l_1, l_2 in the perspective of kinetic energy.

In other words, the system is just a rigid body.